

Sachin Mosambe

AI/ML Engineer

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SUMMARY

AI Engineer specializing in building production-ready LLM applications, agentic AI systems, and AI-powered backend services. M.Tech graduate from IIT Kharagpur with expertise in Python, FastAPI, LangGraph, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), semantic search, vector databases, and AWS. Experienced in developing scalable AI solutions, multi-agent workflows, enterprise integrations, and production-grade applications with a strong focus on reliability, observability, and real-world business impact.

EDUCATION

IITM Pravartak Certified AI and Machine Learning Program CGPA: 10/10	Jul 2025 — Sep 2025
Guvi (HCL) — Comprehensive certification in AI, ML, Deep Learning, NLP, and Generative AI	
Master of Technology (M.Tech) in Infrastructure Design Management CGPA: 9.09/10	Aug 2022 — May 2024
Indian Institute of Technology (IIT) Kharagpur	
Bachelor of Technology (B.Tech) in Civil Engineering CGPA: 8.16/10	Aug 2017 — May 2021
Government College of Engineering Amravati	

SKILLS

Programming: Python, SQL

AI/ML: Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents, Multi-Agent Systems, LangChain, LangGraph, Model Context Protocol (MCP), Semantic Search

Frameworks & Databases: FastAPI, PostgreSQL, MySQL, Redis, Celery, FAISS, Qdrant

Data Science & MLOps: Pandas, NumPy, Scikit-learn, PyTorch, MLflow, Git, CI/CD

Cloud & Tools: AWS (EC2, S3, Bedrock), Docker, LangSmith, Cursor, Windsurf, Claude Code

EXPERIENCE

AI/ML Developer | Aptino Technologies Mar 2026 — Present

- Architected and developed an **AI-native Applicant Tracking System (ATS)** using **Python, FastAPI, PostgreSQL, Redis, Celery, and Docker**, building scalable backend services for resume parsing, candidate evaluation, recruiter workflows, asynchronous processing, and audit logging.
- Designed and implemented **LangGraph-based multi-agent AI systems** with supervisor workflows, **Model Context Protocol (MCP)** integrations, tool-use architectures, **Retrieval-Augmented Generation (RAG)**, semantic search, and **FAISS** vector retrieval to automate resume analysis and intelligent candidate matching.
- Built production-ready **LLM applications** using **LangChain**, implementing prompt engineering, context management, structured outputs, fallback strategies, AI guardrails, and multi-model routing across **Groq, OpenRouter, and AWS Bedrock**.
- Developed an **AI Interview Intelligence Platform** using **FastAPI, React, WebSockets, AWS Transcribe, AWS Polly, and LangGraph**, enabling real-time AI voice interviews, contextual question generation, automated candidate evaluation, and recruiter insights.
- Improved production reliability and engineering velocity through **LangSmith** observability, structured logging, Docker deployments, Celery retry workflows, enterprise API integrations, and AI-native development with **Cursor** and **Windsurf**, delivering scalable, production-ready AI solutions.

PROJECTS

RetrieveIQ – Intelligent Retrieval-Augmented Generation (RAG) System

- Architected a production-grade **Retrieval-Augmented Generation (RAG)** system using **LangChain, FAISS**, and semantic document retrieval, enabling grounded question answering across multiple document sources.
- Improved retrieval quality through **cross-encoder re-ranking**, achieving **82% Top-5 retrieval accuracy** measured using **RAGAS** evaluation metrics, and integrated OpenAI LLMs for citation-backed answer generation.

EMIPredict AI – Financial Risk Assessment Platform

- Processed and engineered features from **400K financial records** using **Pandas** and **NumPy**, performing data cleaning, feature engineering, imputation, and outlier handling to build robust machine learning datasets.
- Developed and optimized **XGBoost** classification models achieving **94.82% accuracy** and **0.9953 ROC-AUC**, and deployed an end-to-end **Streamlit** application with **MLflow** experiment tracking, reducing loan processing time by **80%**.

AI Agent Chatbot – Tool-Augmented Conversational Assistant

- Built a **LangGraph-based** AI agent with tool-use workflows using **LangChain**, integrating OpenAI and Groq LLMs to deliver context-aware conversational assistance with external search capabilities.
- Containerized the application using **Docker** and deployed it with **CI/CD** to Google Cloud Run, enabling scalable, production-ready deployment and interactive web-based user experiences.

CERTIFICATIONS

Machine Learning Specialization (DeepLearning.AI) | Deep Learning Specialization (DeepLearning.AI) | Natural Language Processing Specialization (DeepLearning.AI) | Generative AI with Large Language Models (DeepLearning.AI & AWS) | IITM Pravartak Certified AI & Machine Learning Program (GUVI)

LEADERSHIP & ACHIEVEMENTS

Led 200-participant conference as Core Committee Member at IIT Kharagpur | Mentored 60+ students as Teaching Assistant | Won Doubles Championship in Table Tennis at Kalpana Chawla Trophy, GCOEA (2019)